

Does conditioning leave a structural memory?

From copper evidence to a stainless-steel test

Yinon Ashkenazy and collaborators

HUJI · CERN · Consorzio RFX

MeVArc 2026



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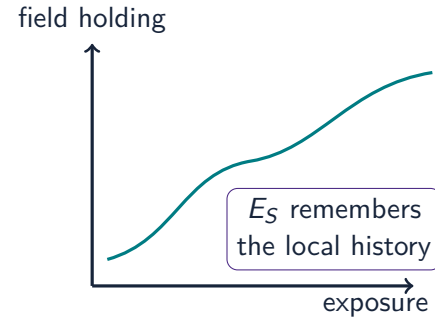


Open with the operational fact: every high-gradient device must be conditioned, but we still do not know what microscopic variable is being changed. The talk has two halves. First, a completed copper result that gives a structural candidate for that variable. Second, a stainless-steel experiment designed to test whether the mechanism generalizes.

Conditioning works—but what is being conditioned?

- ▶ Repeated high-field exposure raises voltage-holding capability.
- ▶ Conditioning follows **pulse count**, not simply breakdown count.
- ▶ A local state variable, E_S , captures the evolving field tolerance.

The missing link is a measurable material state that survives between pulses.



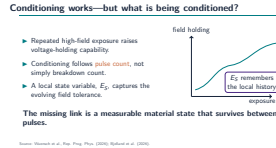
Source: Wuensch et al., Rep. Prog. Phys. (2026); Bjelland et al. (2026).

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Does conditioning leave a structural memory?

└ Conditioning works—but what is being conditioned?

Define E_S operationally: the local field level to which a surface element has conditioned in the Monte Carlo description. Emphasize that it is phenomenological; the model did not identify its microscopic carrier. The question for this talk is whether the subsurface dislocation population can provide that memory.



Dislocations are a plausible carrier of that memory

- Breakdown statistics and material trends track barriers to dislocation motion.
- The MDDF model treats initiation as a collective dislocation instability.
- Local TEM finds a ~200 nm dislocation-denuded layer in conditioned hard Cu.

The hypothesis is cumulative, field-driven plastic rearrangement below macroscopic yield.

Maxwell stress

$$\sigma_M = \frac{\epsilon_0 E^2}{2}$$

at 80 MV/m: 0.028 MPa

tiny per pulse, repeated ~10⁹ times

Source: Engelberg et al., PRL 2018 / PRAB 2019; Jacewicz et al., PRAB 2025.

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└ Dislocations are a plausible carrier of that memory

Do not claim ordinary fatigue equivalence: the Maxwell stress is about three orders below annealed-copper yield. The relevance of very-high-cycle fatigue is conceptual—many sub-yield cycles can accumulate irreversible rearrangement. The MDDF model is valuable because it is threshold-free and connects collective fluctuations to the steep field dependence of breakdown.

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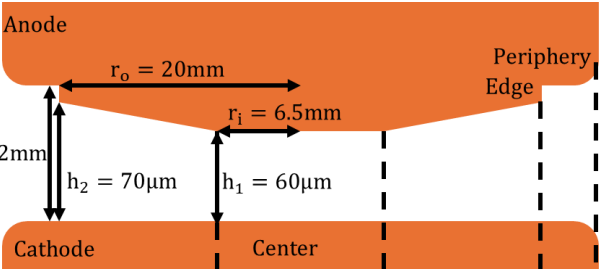
Maxwell stress

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tiny per pulse, repeated ~10⁹ times

One cathode turns field exposure into a controlled coordinate



- OFE Cu, pulsed DC
- 1 μs pulses at 1 kHz
- Peak field $\sim 80\text{ MV/m}$
- Nine EBSD regions: center, edge, periphery, and external reference

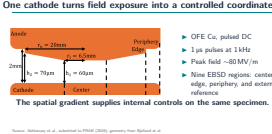
The spatial gradient supplies internal controls on the same specimen.

Source: Ashkenazy et al., submitted to PRAB (2026); geometry from Bjelland et al.

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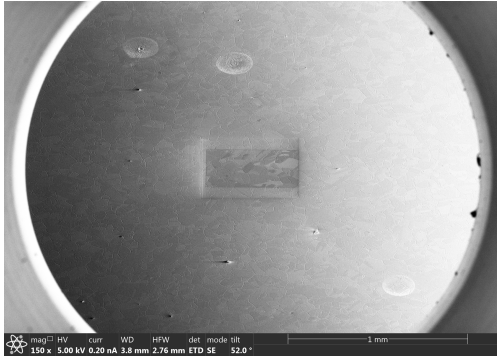
Does conditioning leave a structural memory?

One cathode turns field exposure into a controlled coordinate



This geometry is the core experimental design. The flat center is at the maximum field; the sloped gap creates a continuous radial decay; beyond the anode radius the field is below about 2.5 MV/m. Comparing positions on one cathode reduces sensitivity to cathode-to-cathode preparation differences. A separate identically treated, unexposed cathode gives an external reference.

EBSD maps plastic activity far from breakdown craters



- ▶ 500 μm -wide maps; 3 μm step
- ▶ ~24,000 indexed points per ROI
- ▶ LAM, KAM, and LOS probe intragrain orientation gradients
- ▶ Misorientation is a proxy for geometrically necessary dislocation density

Cleaned ROI is hundreds of microns from visible craters.

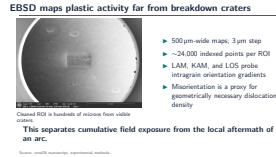
This separates cumulative field exposure from the local aftermath of an arc.

Source: cond26 manuscript, experimental methods.

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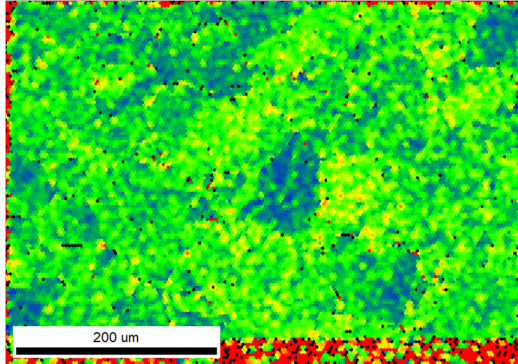
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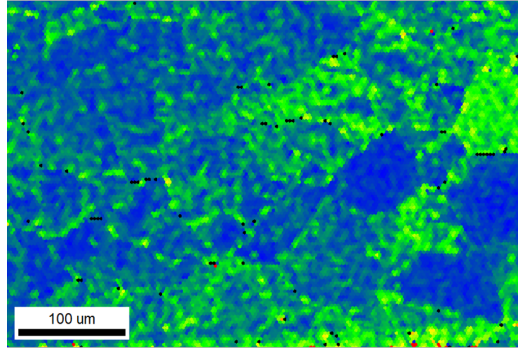


Explain the cleaning and selection briefly: a shallow FIB removal of oxide, less than 10 nm by AFM, followed by EBSD. All regions were deliberately chosen away from craters. LAM is the primary visual metric; two other definitions are used to check that the result is not an artifact of one analysis choice.

High-field copper contains more intragrain curvature



Field-exposed center



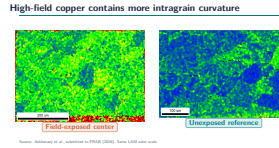
Unexposed reference

Source: Ashkenazy et al., submitted to PRAB (2026). Same LAM color scale.

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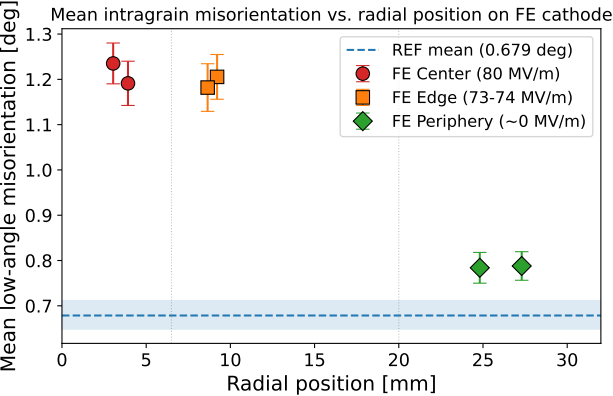
Does conditioning leave a structural memory?

└ High-field copper contains more intragrain curvature



Invite the audience to look at the spatial texture, not just the color average. The field-exposed region carries more high-misorientation structure within grains. State that the comparison shown is representative, while the quantitative conclusion is based on all nine ROIs.

The signal separates into three exposure tiers



- ▶ Center and edge: $\sim 1.2^\circ$
- ▶ External reference: $\sim 0.68^\circ$
- ▶ High field is $\sim 75\%$ higher
- ▶ Grain size does not follow this hierarchy

The ordering follows field history, not merely sample position or grain size.

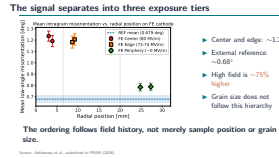
Source: Ashkenazy et al., submitted to PRAB (2026).

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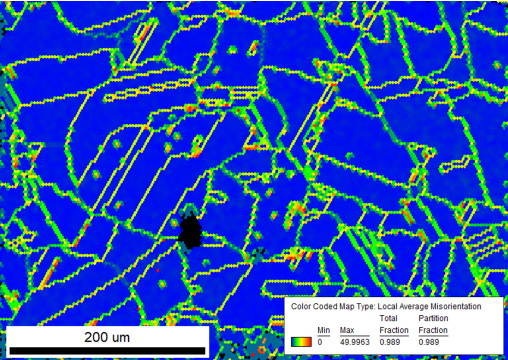
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The signal separates into three exposure tiers

Walk through the tiers: high-field center and edge are similar; the field-exposed periphery is intermediate; unexposed references are lowest. The periphery is useful because it shared machining, heat treatment, mounting, vacuum, and handling with the high-field regions but experienced negligible field. Be explicit that the 75 percent value characterizes this cathode and history, not a universal material constant.



Field exposure populates a high-misorientation tail



LAM tail above 2°

~0.14 field-exposed

~0.016 reference

~8× increase

All center/edge distributions

differ from unexposed references:

$D_{KS} > 0.25$.

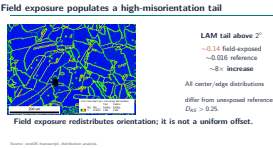
Field exposure redistributes orientation; it is not a uniform offset.

Source: cond26 manuscript, distribution analysis.

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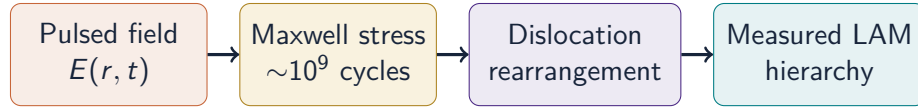
Does conditioning leave a structural memory?

Field exposure populates a high-misorientation tail



The mean is not the whole story. The distribution broadens strongly and acquires a long tail. Mention that LAM, KAM, and LOS reproduce the ordering, and that KS, chi-squared, and Anderson–Darling tests all distinguish the high-field and unexposed distributions. Avoid over-selling pixel-level independence; the statistics demonstrate robust distribution separation across the maps.

A material state now shadows the conditioning state E_S



E_S : phenomenological memory



Misorientation: structural memory

The evolving subsurface dislocation population is a candidate physical basis for E_S .

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Does conditioning leave a structural memory?

└ A material state now shadows the conditioning state E_S

A material state now shadows the conditioning state E_S



Use candidate, not identification. The spatial ordering matches, but a point-by-point quantitative correlation with the simulated $E_S(r)$ still requires more radial positions and finer EBSD. The result is nevertheless the first large-area comparison that connects conditioned and unconditioned regions through a dislocation-sensitive structural observable.

Copper establishes the effect; it does not establish universality

What is established

- ▶ Large-area structural difference
- ▶ Three independent metrics
- ▶ Internal and external controls
- ▶ Spatial ordering consistent with E_S

A stronger test changes both the material and the loading history.

What remains open

- ▶ Full depth profile
- ▶ Grain-orientation dependence
- ▶ More conditioned cathodes
- ▶ Cross-material generality

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A stronger test changes both the material and the loading history.

This is the transition slide. Acknowledge the single conditioned cathode and single external reference. Depth is especially important because TEM reports depletion in the top 200 nm while EBSD reports accumulated curvature over a larger interaction volume. These may be parts of one redistribution profile. The stainless-steel collaboration supplies an orthogonal test rather than a simple replication.

Stainless steel is a deliberately difficult test

Copper
FCC, higher SFE
easier cross-slip
pulsed DC loading
 $\sim 10^9$ cycles

$\neq$

AISI 304 SS
metastable austenite, lower SFE
planar slip / possible twinning
continuous DC exposure
possible $\gamma \rightarrow \epsilon \rightarrow \alpha'$ transformation

A correlated SS response would support a mechanism broader than one alloy or pulse protocol.

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Does conditioning leave a structural memory?

└ Stainless steel is a deliberately difficult test

Clarify that the drawing specifies AISI 304; the exact sub-grade and composition certificate remain outstanding. Stainless steel changes stacking-fault energy, dislocation mobility, and introduces metastable phase transformations. Continuous DC is creep-like rather than the copper experiment's cyclic loading. Similar spatial behavior under these differences would be powerful; different behavior would also be physically informative.

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AISI 304 SS

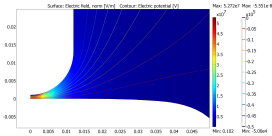
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A correlated SS response would support a mechanism broader than one alloy or pulse protocol.

The RFX electrodes provide a second field gradient



HVPTF sphere-plane assembly



Simulated field map

- ▶ Ø24 mm hemispherical cathode
- ▶ 1 mm gap
- ▶ two exposed electrodes
- ▶ ≥ 100 h at 60–80 MV/m
- ▶ two batch-matched references

Apex-to-periphery variation enables an internal field-correlation test.

Source: Consorzio RFX electrode drawing and electrostatic simulation; RFX project status, July 2026.

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Does conditioning leave a structural memory?

└ The RFX electrodes provide a second field gradient

The RFX High Voltage Padova Test Facility operates a long-gap DC system. The local field is highest at the hemisphere apex and falls toward the periphery. That gradient is crucial because it can distinguish a conditioning response from sample-to-sample electropolishing differences. Conditioning histories and a numerical field-versus-angle vector are still requested; do not imply that the full exposure map is already available.

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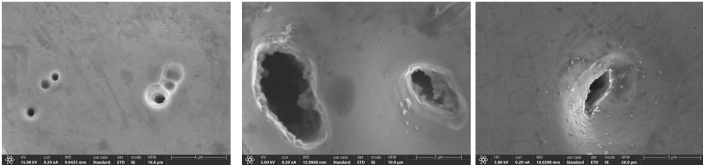
Source: Consorzio RFX electrode drawing and electrostatic simulation; RFX project status, July 2026.

The first SEM survey reveals a new surface signature

Sample 1-25 ref

Sample 3-25 tested

Sample 4-25 tested



- ▶ Reference: cleaner, rounded cavities
- ▶ Tested: irregular erupted sites
- ▶ Typical scale: 1–7 μm
- ▶ No clear molten rim or recast layer

The morphology is consistent with ruptured near-surface inclusions—not yet proof of a field mechanism.

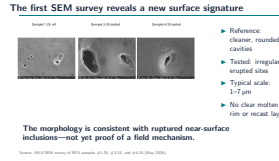
Source: HUJI SEM survey of RFX samples #1-25, #3-25, and #4-25 (May 2026).

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Does conditioning leave a structural memory?

└ The first SEM survey reveals a new surface signature

Be conservative. Chemistry is dominated by elements expected in native 304 inclusion families: Ca, Al, Mg, Si, Mn, Cr, Ti, and V; potassium is a contamination flag and carbon is ambiguous. The eruption morphology lacks the molten rim and recast signature of a conventional arc crater. The decisive test is a FIB cross-section through erupted and intact inclusions, plus a quantitative inclusion census on exposed and reference areas.



Early EBSD behavior may already be diagnostic

Reference #1-25

- ▶ good diffraction patterns
- ▶ LAM and grain statistics extracted
- ▶ Cu analysis pipeline transfers

Field-exposed #4-25

- ▶ poor pattern quality
- ▶ higher current / longer exposure did not recover it
- ▶ cause is unresolved

Three hypotheses: preparation variation conditioning-induced defects
oxide/contamination overlayer

Pattern degradation is evidence only if it repeats and follows the field gradient.

Source: RFX project status and HUJI EBSD update, 13 July 2026.

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Does conditioning leave a structural memory?

└ Early EBSD behavior may already be diagnostic

This is deliberately framed as an observation, not a result. EBSD information depth is tens of nanometers, so defects or a surface overlayer could degrade pattern quality. A repeat on the second exposed and second reference sample is essential. Quantitative image-quality metrics, hit rate, mean angular deviation, and raw EBSPs should be retained.

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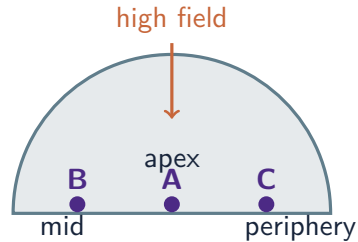
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The decisive comparison is within each exposed hemisphere



Conditioning signature

Pattern quality and/or misorientation vary monotonically with local field on both exposed samples.

Preparation signature

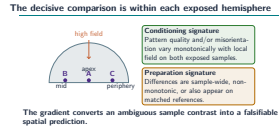
Differences are sample-wide, non-monotonic, or also appear on matched references.

The gradient converts an ambiguous sample contrast into a falsifiable spatial prediction.

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Does conditioning leave a structural memory?

└ The decisive comparison is within each exposed hemisphere



Lay out the measurement matrix: apex, mid-latitude, and periphery on both exposed electrodes, with matched positions on both references. Add EDS carbon/oxygen checks to separate defect density from an overlayer. Correlate quantitative EBSD pattern-quality metrics first; only compare LAM/KAM where indexing quality is adequate and controlled.

Depth profiling can manufacture the signal we seek

Ga damage A 30 kV FIB

can damage a layer comparable to the EBSD information depth.

Redeposition Serial

milling can mix material and obscure the true depth coordinate.

False martensite Ion

irradiation can transform metastable austenite to BCC, mimicking α' formation.

Pilot-first control: identical preparation on exposed and reference material; low-energy final polish; dose bookkeeping; go/no-go before the full $\sim 1\ \mu\text{m}$ campaign.

Preparation artifacts are part of the experiment, not a post-processing detail.

Source: RFX artifact review; Babu et al., Acta Mater. 120, 391 (2016).

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Does conditioning leave a structural memory?

└─Depth profiling can manufacture the signal we seek

This is the main technical risk. A ten-nanometer serial step is not physically meaningful if the ion-damaged layer is 20–30 nm and the EBSD signal depth is 20–50 nm. Metastable austenite adds a special hazard: Ga or Xe irradiation can generate BCC material with orientation relations that look like deformation martensite. The pilot must establish the artifact floor on a reference before any phase-change claim.

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Every outcome constrains the conditioning mechanism

SS tracks field

Supports a cross-material structural state despite different slip physics and loading.

SS changes differently

Separates material-specific carriers: planar slip, twins, phase change, or inclusions.

No resolved change

Bounds the accessible depth, dose, and sensitivity of dislocation-based models.

The next result is not simply “positive or negative”; it is a mechanism discriminator.

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Does conditioning leave a structural memory?

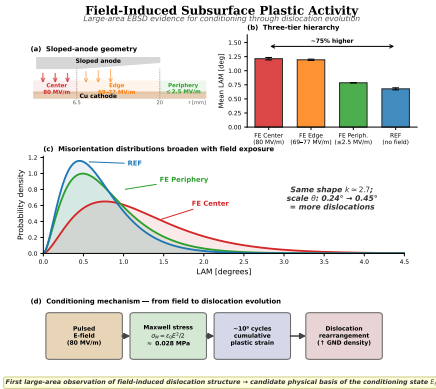
└ Every outcome constrains the conditioning mechanism

This slide is the payoff for presenting RFX as a future direction. A positive correlation supports generality. A qualitatively different response can identify which material degrees of freedom store the state. A null result, if exposure and measurement sensitivity are quantified, places a useful bound and may distinguish cyclic from continuous loading.

Every outcome constrains the conditioning mechanism		
SS tracks field Supports a cross-material structural state despite different slip physics and loading.	SS changes differently Separates material-specific carriers: planar slip, twins, phase change, or inclusions.	No resolved change Bounds the accessible depth, dose, and sensitivity of dislocation-based models.
The next result is not simply “positive or negative”; it is a mechanism discriminator.		

Conditioning appears to leave a subsurface memory

- 1. In copper, large-area misorientation is ~75% higher in high-field regions.
- 2. Its spatial hierarchy follows the conditioning-state profile E_S .
- 3. Stainless steel tests whether that memory survives a change of alloy and loading mode.



We can now ask what conditioning stores—and where it stores it.

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Does conditioning leave a structural memory?

Conditioning appears to leave a subsurface memory

Return to the opening question. The copper data provide a measurable structural memory correlated with field exposure and the modeled conditioning state. The RFX experiment is designed to falsify or generalize that interpretation. End on the two coordinates now accessible: lateral field gradients and depth-resolved structure. Invite discussion about the most discriminating SS measurements and the connection to operational conditioning models.

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Backup: RFX sample and data status

Sample	Role	Surface	Current status
#1-25	reference	electropolished	SEM + good EBSD
#2-25	reference	electropolished	follow-up acquisition planned
#3-25	field-exposed	≥ 100 h HV	SEM; EBSD follow-up
#4-25	field-exposed	≥ 100 h HV	SEM; poor initial EBSD

- ▶ Outstanding: full conditioning report; numerical $E(\theta)$ vector; alloy certificate; electropolishing protocol.
- ▶ Needed for interpretation: quantitative EBSD quality metrics and raw patterns; matched EDS overlay checks.

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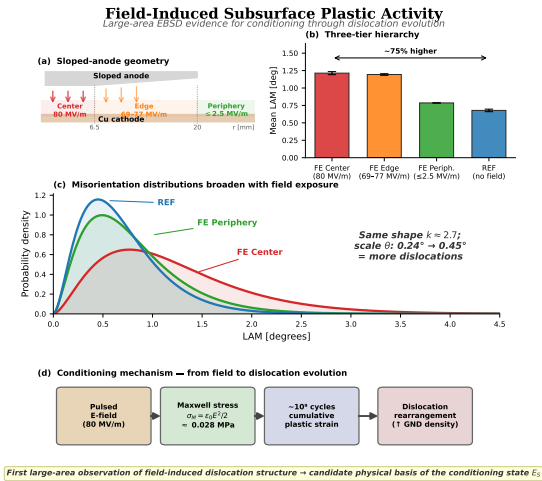
└ Backup: RFX sample and data status

Use only if asked about readiness or missing inputs. This status reflects the written project record through 13 July 2026.

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Backup: copper result in one figure



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Does conditioning leave a structural memory?

Backup: copper result in one figure

Use this as a compact answer to questions about geometry, distributions, or the proposed conditioning mechanism.

